

Advanced Econometrics Lecture 01

dr Marcin Chlebus, dr Rafał Woźniak

Faculty of Economic Sciences, University of Warsaw

Warsaw

Plan of the course

No.	Topic
1.	Intro / Panel data models
2.	Panel data models / Intro to Maximum Likelihood Method
3.	Binary dependent variables
4.	Ordered Logit & Probit
5.	Multinomial Logit, Conditional Logit
6.	Models for count data
7.	Censored data, sample selection, censored data model (tobit)
8.	General-to-Specific Approach
9.	Introduction to Time Series, Spurious regression
10.	Stationarity and Nonstationarity
11.	DL and ARDL models
12.	ARMA and ARIMA models
13.	Cointegration analysis
14.	Instrumental variables
15.	Advanced Panel Models

Assessment of the course

- ① Short tests (20%)
- ② Empirical study (40%)
 - Empirical data analysis using the appropriate method discussed during the course or related.
 - Word report and codes in R.
 - Deadline 31.05.2026 19:59
- ③ Final exam (40%)
 - You can bring one handwritten A4 sheet of paper with any notes you want.
 - 3 empirical problems to be solved (with several subpoints).
- ④ The final exam has to be passed.
- ⑤ The project has to be passed.
- ⑥ Lab class has to be passed to take the exam.

Final grades

Final grade

- a) 20% short tests
- b) 40% project
- c) 40% exam

You have to:

- a) Pass the project
- b) Pass the exam

Grade

- $< 50\%$ - 2
- $[50\%, 60\%)$ - 3
- $[60\%, 70\%)$ - 3.5
- $[70\%, 80\%)$ - 4
- $[80\%, 90\%)$ - 4.5
- $[90\%, 100\%]$ - 5

Exam

- 90 minutes
- 3 exercises with multiple subquestions
- You have to pass it
- Grades as above

Recommended literature

- ① J. Wooldridge, *Introductory Econometrics, A Modern Approach*, Cengage
- ② Florian Heiss and Daniel Brunner, *Using R, Python and Julia for Introductory Econometrics*, (link)
- ③ W.H. Greene, *Econometric Analysis*, Pearson
- ④ W. Enders, *Applied Econometric Time Series*, Wiley.
- ⑤ W.W. Charemza, D.F. Deadman, *New Directions in Econometric Practice*, Edward Elgar Publishing
- ⑥ R. Davidson, J.G. McKinnon, *Estimation and Inference in Econometrics*, Oxford University Press
- ⑦ G.S. Maddala, *Limited-Dependent and Qualitative Variables in Econometrics*, Cambridge University Press
- ⑧ R.C. Hill, W.E. Griffiths, G.C. Lim, *Principles of Econometrics*, Wiley
 - C. Colonescu, *Using R for Principles of Econometrics*
 - Free online <https://bookdown.org/ccolonescu/RPoE4/>

Suppose your superior asks you to estimate an econometric model for the following topics. Which model should be chosen from the list?
ARIMA, ARDL, Logit/Probit, Ordered logit/probit, Unordered logit/probit, Panel model, Tobit, Error-Correction Mechanism (Cointegration).

Let's consider the households food expenditures model

Type I

$$food_i = 463.95 + 0.079 \text{ income}_i + e_i \quad (1)$$

Type II

$$\ln(food_i) = 3.697 + 0.357 \ln(income_i) + e_i \quad (2)$$

$$\ln(food_i) = 3.630 + 0.349 \ln(income_i) + 0.01 \ln(kids_i) + e_i \quad (3)$$

Type III

$$\ln(food_i) = 3.630 + 0.349 \ln(income_i) + 0.114 kids_i + e_i \quad (4)$$

Type IV Let's consider the model for unemployment rate of the form

$$unmpl_t = 0.011 - 0.247 \text{ gdp}_t - 0.310 \text{ cpi}_t + e_t \quad (5)$$

where *unmpl* denotes unemployment rate, *gdp* – GDP growth rate, and *cpi* stands for consumer price index, i.e. inflation rate.

Military expenses

Let's consider the following model for military expenses among countries

$$\begin{aligned} \ln(ARMY_i) &= 1.284 + 0.005 \ln(GDP_i) - 0.01 \ln(POPULATION_i) \\ &- 1.10 noSEA_i + \varepsilon_i \end{aligned} \quad (6)$$

where *ARMY* denotes military budget in millions of USD, *GDP* stands for Gross Domestic Product in millions of USD, and *noSEA* is a dummy variable equal to 1 if a country has no direct access to international waters.

Introduction

1 Data

- a) cross-sectional
- b) time series
- c) panel data

2 Variables

- a) continuous
- b) discrete
- c) qualitative
- d) quantitative

Example

What are the determinants of household medical expenditures?

What does panel data look like?

firm	year	prod	area	labor	fert
1	1990	7.87	2.5	160	207.5
1	1991	7.18	2.5	138	295.5
1	1992	8.92	2.5	140	362.5
1	1993	7.31	2.5	127	338
1	1994	7.54	2.5	145	337.5
2	1990	10.35	3.8	184	303.5
2	1991	10.21	3.8	151	206
2	1992	13.29	3.8	185	374.5
2	1993	18.58	3.8	262	421
2	1994	17.07	3.8	174	595.7
3	1990	9.98	3.4	170	252
3	1991	10.44	4.3	148	444
3	1992	10.4	4.3	198	283.5
3	1993	8.92	4.3	189	287
3	1994	7.36	4.3	214	294
⋮	⋮	⋮	⋮	⋮	⋮